

To address this requirement, Microsoft has introduced *cryptlets* as part of Blockchain 3.0, defined as blockchain of data, logic and cloud services.

Azure Blockchain Service

Azure Blockchain Service is a fully managed ledger service that gives users the ability to grow and operate blockchain networks at scale in Azure, via unified control for both infrastructure management as well as blockchain network governance.

It provides the following:

- Simple network deployment and operations
- Consortium management
- Smart contracts with development tools

It provides support for the Ethereum Quorum ledger using the Istanbul Byzantine Fault Tolerance (IBFT) consensus mechanism.

Features of Azure Blockchain Service

Azure network deployment and operations: Azure Blockchain Service is developed through the Azure CLI, Visual Studio Code, or the Azure portal with the assistance of the Azure blockchain extension. Network deployment is made simple in the Azure Blockchain Service. It includes provisioning both the validator and transaction nodes, as well as Azure virtual networks for service-managed storage and security isolation. When a new member is deployed, users join or create a consortium. These consortiums allow multiple parties in several Azure subscriptions to communicate securely with each other on a shared blockchain. This simplifies blockchain network deployment to a great extent and reduces the time from days to minutes.

Performance and service tiers: The two service tiers offered in Azure Blockchain Service are basic and

standard. Each tier offers several capabilities to support lightweight development, and also allows the testing of workloads to massively scaled production blockchain deployments. Both these tiers include two standard validator nodes, one basic validator node, and one transaction node.

Maintenance and security: Once you provision your first blockchain member, you will be able to add additional transaction nodes to it. These nodes need configuration for access and are safeguarded through firewall rules. Other options that can be used for securing transaction node access are access keys, basic authentication and Azure Active Directory integration.

Monitoring and logging: The Azure service offers rich metrics through the Azure Monitor Service that provides insights into the memory, storage usage and CPU of the nodes. Apart from these, it also provides insights into blockchain network activity such as transaction queue depth, active connections and transactions, and the blocks mined. Azure makes it possible to customise metrics to provide the insights. It is also possible to define thresholds through alerts to help users trigger actions such as sending an email or text message, or running a logic app or Azure function. Users can also view logs related to the Quorum ledger using Azure Log Analytics.

Built-in consortium management: As mentioned earlier, at the time of deploying the first blockchain member, you will either join or create a consortium. A consortium can be understood as a logical group that manages the connectivity and governance between blockchain members who transact in a multi-party process. Azure offers governance controls through pre-defined smart contracts that determine the actions members

present in the consortium are allowed to take. The administrator can customise the governance controls as and when needed. You must have an invite to join a consortium. Consortium management actions like removing and adding members from a consortium can be accessed through a REST API and power shell.

Architecture of the Azure blockchain solution

Azure blockchain provides two solution architectures.

Supply chain track and trace:

Supply chain is one of the most common problems that can be solved using blockchain. The main idea is to assure that an asset can move through a network without modification during transit. The architecture of the supply chain using Azure can be viewed in Figure 1.

Blockchain workflow

application: This helps blockchain digitise the workflow, and improves both the internal and external functioning of the organisation. It covers most of the blockchain apps' anatomy, and assists the developer in implementing diverse use cases. Figure 2 highlights the blockchain workflow application.

Azure blockchain development kit

Microsoft Azure blockchain development kit is built on Microsoft serverless technologies and integrates Microsoft and third party SaaS. This kit extends the capabilities of blockchain developer templates and Azure blockchain workbench. It incorporates Azure services for key management, off-chain identity and data, as well as monitoring and messaging APIs into reference architecture to design blockchain based applications.

The Microsoft Azure blockchain development kit has the following features: